

Lesson 2 · Turning Points and Optimisation

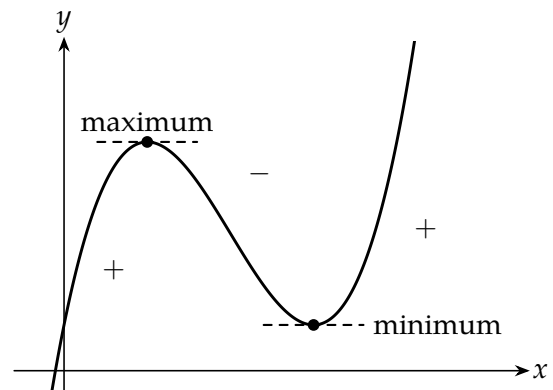
NCEA Level 2 Mathematics · AS 91262 Apply calculus methods in solving problems · Handout

1 The one idea

At the top of a hill or the bottom of a valley the tangent is flat.

At a turning point the gradient is zero: $f'(x) = 0$.
Maximum or minimum? Look at the gradient on each side.

On the left of a **maximum** the curve goes up (+), on the right it goes down (−). A **minimum** is the other way round: − then +.



2 Finding turning points

Write it in this order

1. Differentiate: $f'(x)$.
2. Set $f'(x) = 0$ and solve for x (factorise, formula, or EQUA).
3. Find each y from $f(x)$, not from $f'(x)$.
4. Decide maximum or minimum, and say why (section 3).

Worked example 1

Find the turning points of $f(x) = x^3 - 6x^2 + 9x + 1$ and state their nature.

Step 1 $f'(x) = 3x^2 - 12x + 9 = 3(x^2 - 4x + 3) = 3(x - 1)(x - 3)$

Step 2 $f'(x) = 0$ when $x = 1$ or $x = 3$.

Step 3 $f(1) = 1 - 6 + 9 + 1 = 5$ and $f(3) = 27 - 54 + 27 + 1 = 1$.

Step 4 $f''(x) = 6x - 12$: $f''(1) = -6 < 0$, so $(1, 5)$ is a **maximum**; $f''(3) = 6 > 0$, so $(3, 1)$ is a **minimum**.

3 Maximum or minimum? Two tests

test	what you see	conclusion
gradient each side	f' goes $+$ $\rightarrow$ 0 $\rightarrow$ $-$	maximum
	f' goes $-$ $\rightarrow$ 0 $\rightarrow$ $+$	minimum
second derivative	$f''(x) < 0$ at the point	maximum (curve bends down)
	$f''(x) > 0$ at the point	minimum (curve bends up)

$f''(x)$ means “differentiate twice”; it is also written $\frac{d^2y}{dx^2}$. For the gradient test, choose x -values just either side and *show* the substitution: e.g. $f'(0) = 9 > 0$ and $f'(2) = -3 < 0$ around $x = 1$.

Sentence frame: “ $f''(1) = -6 < 0$, so the turning point at $(1, 5)$ is a maximum.”

4 Increasing and decreasing

Increasing where $f'(x) > 0$. **Decreasing** where $f'(x) < 0$.

Find where $f'(x) = 0$ first; these x -values split the line into intervals. Then test each interval. If $f'(x)$ is an up-parabola, it is **negative between its roots**.

Worked example 2

For $f(x) = x^3 - 6x^2 + 9x + 1$, find where f is decreasing.

Step 1 $f'(x) = 3(x - 1)(x - 3)$, zero at $x = 1$ and $x = 3$.

Step 2 Up-parabola, so negative between the roots: $1 < x < 3$.

Increasing for $x < 1$ or $x > 3$. Matches the picture: down between the maximum and the minimum.

Worked example 3

For which x is $y = 5 + 4x - x^2$ increasing?

Step 1 $\frac{dy}{dx} = 4 - 2x$

Step 2 $4 - 2x > 0$ gives $2x < 4$, so $x < 2$.

5 Optimisation: the biggest or smallest value

Write it in this order

1. Write the quantity as a formula in **one variable** (use the constraint to remove the other).
2. Differentiate.
3. Set the derivative = 0 and solve.
4. Justify maximum or minimum (f'' or the gradient each side).
5. Answer the question asked, with units.

Worked example 4 · formula given

A ball's height is $h = 2 + 19.6t - 4.9t^2$ metres after t seconds. Find the maximum height.

Step 1 $\frac{dh}{dt} = 19.6 - 9.8t = 0$, so $t = 2$ s.

Step 2 $h = 2 + 19.6(2) - 4.9(2)^2 = 2 + 39.2 - 19.6 = 21.6$

Step 3 $\frac{d^2h}{dt^2} = -9.8 < 0$, so this is a maximum. **Maximum height 21.6 m.**

Worked example 5 · build the formula (Merit/Excellence)

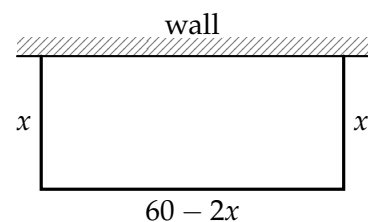
A farmer has 60 m of fence for a rectangular pen against a wall (three sides fenced). Find the maximum area.

Step 1 Sides x, x and $60 - 2x$: $A = x(60 - 2x) = 60x - 2x^2$

Step 2 $\frac{dA}{dx} = 60 - 4x = 0$, so $x = 15$.

Step 3 $\frac{d^2A}{dx^2} = -4 < 0$, so maximum.

Step 4 $A = 15 \times 30 = 450 \text{ m}^2$ (pen 15 m by 30 m).



6 What each grade looks like here

Achieved	find a turning point, or the maximum of a given formula, with the method shown.
Merit	link steps: justify the nature, find an interval, find a constant from a turning point, interpret in context.
Excellence	build the model from words and a diagram, optimise it, justify, and answer in context; or show a general result with letters.

7 Checking with the fx-9750GIII

- GRAPH: enter Y1, DRAW, then G-SOLVE (SHIFT F5) → MAX or MIN. Use it to check a turning point, never as the method.
- EQUA → POLY solves $f'(x) = 0$ when f is a quadratic.

8 Ways to lose marks

From the material plus teaching observation.

- L1 Stopping at x : a turning point is a point, so find y from $f(x)$ as well.
- L2 Writing “maximum” with no reason. Justify with $f''(x)$ or the gradient on each side.
- L3 Increasing or decreasing: giving the turning points instead of an interval, or the inequality the wrong way round.
- L4 Differentiating a formula that still has two variables. Use the constraint to get one variable first.
- L5 Answering the wrong quantity: the question asks for the maximum area, not the x that gives it.
- L6 No units or context in the final sentence (m, s, \$, people per hour).